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DEPARTMENT OF
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CERTIFICATE OF APPRECIATION

This certificate recognizes

Mr. Mohamed Harees M

from

E.G.S. Pillay Engineering College, Nagapattinam

for his exemplary achievement as the recipient of the
Best Oral Presentation Prize at the International
Conference on Health and Disease Management
(ICHDM 2026), held on February 26 and 27, 2026. His
outstanding scholarship, excellence in presentation,
and significant contributions to the advancement of
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T904

Eco-Friendly Synthesis of Copper Oxide Nanoparticles from *Leucas Aspera* Extract for Faster Antimicrobial Wound Healing

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Abstract

This study explores the green synthesis of copper oxide nanoparticles (CuO NPs) utilizing toluene-soluble *Leucas Aspera* leaf extract as a stabilizing agent, aiming to develop sustainable antimicrobial textiles for biomedical use. Comprehensive characterization techniques, including UV-Visible spectroscopy, X-ray diffraction (XRD), scanning electron microscopy (SEM), transmission electron microscopy (TEM), and energy-dispersive X-ray analysis (EDAX), confirmed the successful formation and stabilization of CuO NPs. The synthesized nanoparticles were applied to colored cotton fabrics, resulting in uniform nanoparticle adsorption and significant enhancement of antibacterial properties. Antimicrobial assays against pathogens such as *Staphylococcus*, *E. coli*, *Corynebacterium*, *Micrococcus*, and *Brevibacterium* demonstrated strong and sustained antibacterial activity, with inhibition zones up to 28 mm. The synergistic effect of bioactive compounds from *Leucas Aspera* extract contributed further to the antimicrobial efficacy, while SEM imaging revealed effective nanoparticle distribution without compromising fabric porosity or flexibility. The modified cotton's improved resistance to microbial colonization and its potential to promote tissue regeneration underscore its suitability for wound healing applications. This research highlights the promise of green nanotechnology in developing next-generation wound care materials. Future directions include optimizing nanoparticle concentration and release, assessing in vivo efficacy, and integrating smart, responsive features for personalized healthcare. The approach also offers scalability for industrial production and adaptability for various antimicrobial medical textiles, providing an eco-friendly, cost-effective solution for infection control and advanced wound management.

Keywords: Eco-friendly synthesis, Copper oxide nanoparticles, *Leucas aspera* leaf extract, Green synthesis, Antimicrobial cotton, Wound healing.

